

WHAM-XREX-PFMG474 Test Campaign Report

10 s ramp / 5 s flat-top

Generated from the 2026-09-21 controller × simulated-Transrex campaign

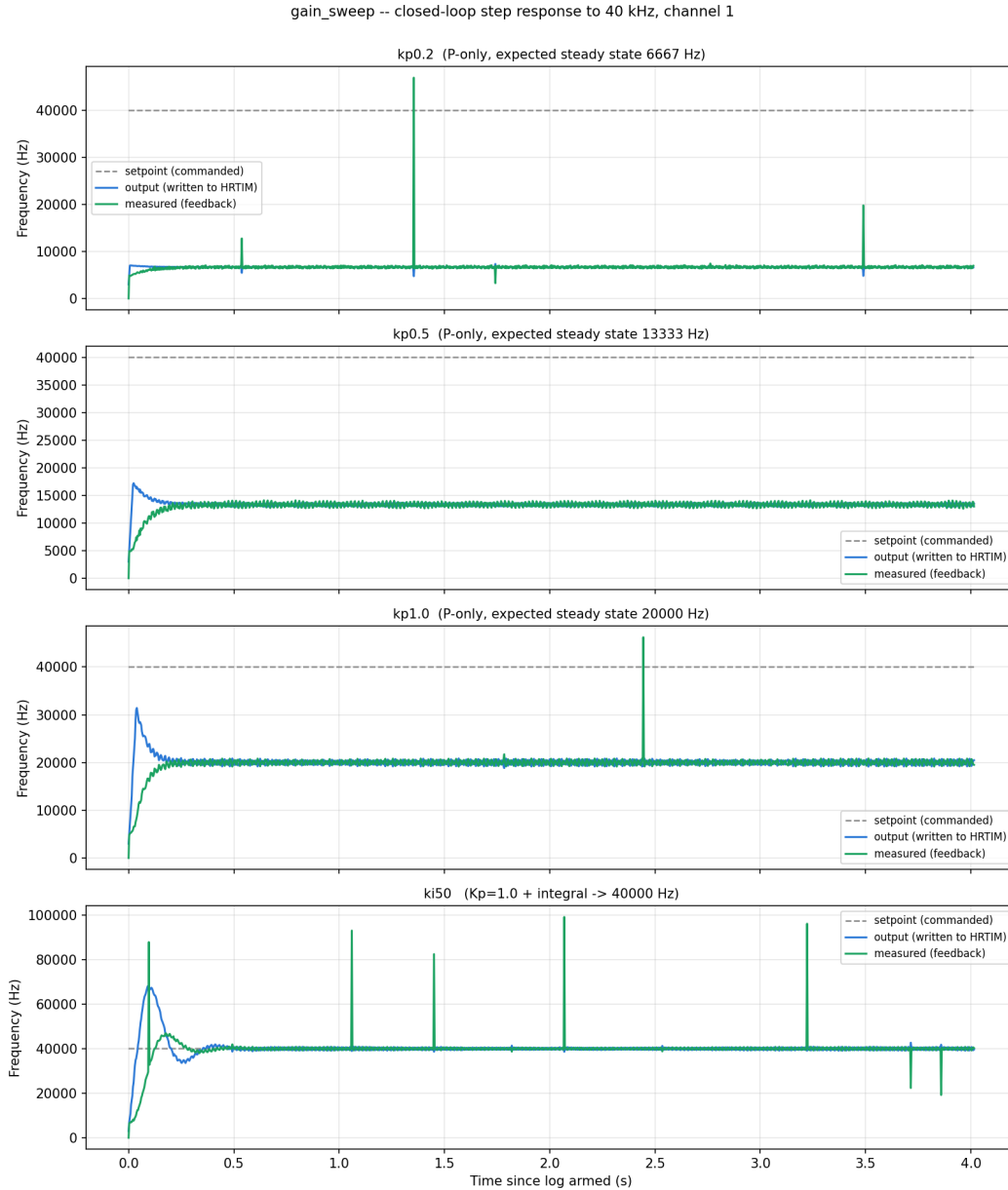
September 21, 2026

Summary

Eight tests were run against the cross-wired controller/simulator rig (firmware d74b0653-dirty, 1750-sample PID log). **6 of 7 discrete scenarios passed** and the **45-shot matrix passed 45/45**. The single non-pass is one criterion-sensitive point in the gain sweep (see §1), not a control defect. Each test below states what was run, why it passed or failed, and shows its 4-panel plot.

1 Gain sweep — closed-loop PID ($P + I$) — 5/6 PASS

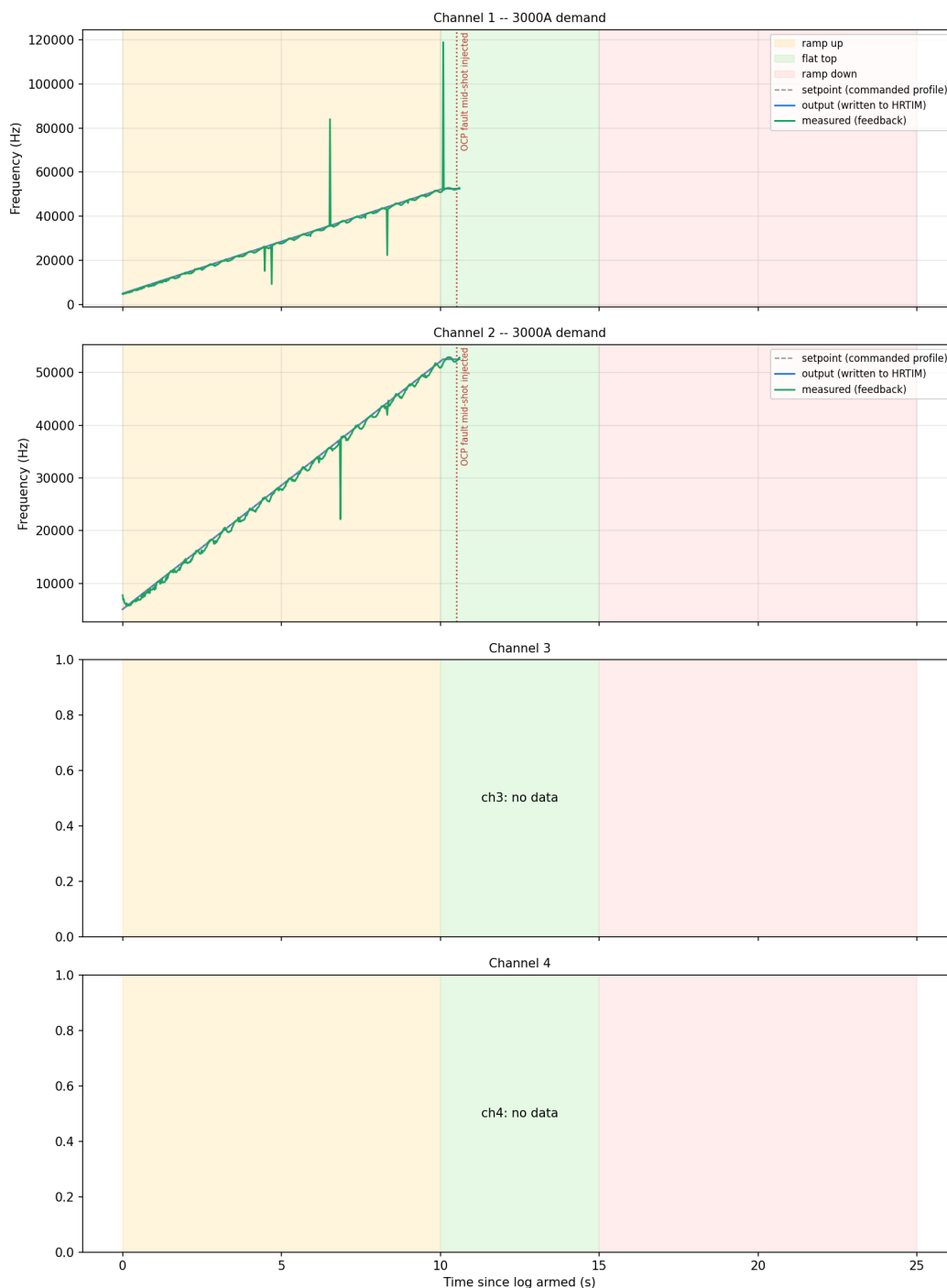
This test exercised the closed-loop PID controller against the simulated Transrex plant by commanding a 40 kHz step on channel 1 under six gain sets: three P-only (K_p 0.2 / 0.5 / 1.0, $K_i = K_d = 0$) and three PI ($K_p = 1.0$, K_i 10 / 50 / 100, $K_d = 0$). The P-only runs **passed** because the measured frequency settled at the theoretical pure-P steady-state value $K_p \cdot 40000 / (1 + K_p)$ — i.e. the expected steady-state error when there is no integral action. The PI runs **passed** because integral action eliminated that error, all three converging to within ~ 550 Hz of 40 kHz (40 236 / 40 255 / 40 543 Hz). The one **FAIL** — ki100 — is a verdict-criterion artifact rather than a control problem: its final sample landed 43 Hz outside the ± 500 Hz tolerance while clearly converged, an artifact of judging by a single ringing-phase sample.



2 Overcurrent fault injection (ocp) — PASS

Two channels were gated open-loop (ramp 10 s / flat 5 s, 3000 A) and an overcurrent fault was injected on channel 1 ~ 0.5 s into the flat top. The test **passed** because STATE? reported OVERCURRENT 1, the faulted channel hard-disabled immediately, and the surviving channel 2 kept running at a derated 21 660 Hz — exactly the specified per-channel overcurrent response (immediate disable for the faulted channel, a $1/N$ step-down then ramp for the survivors).

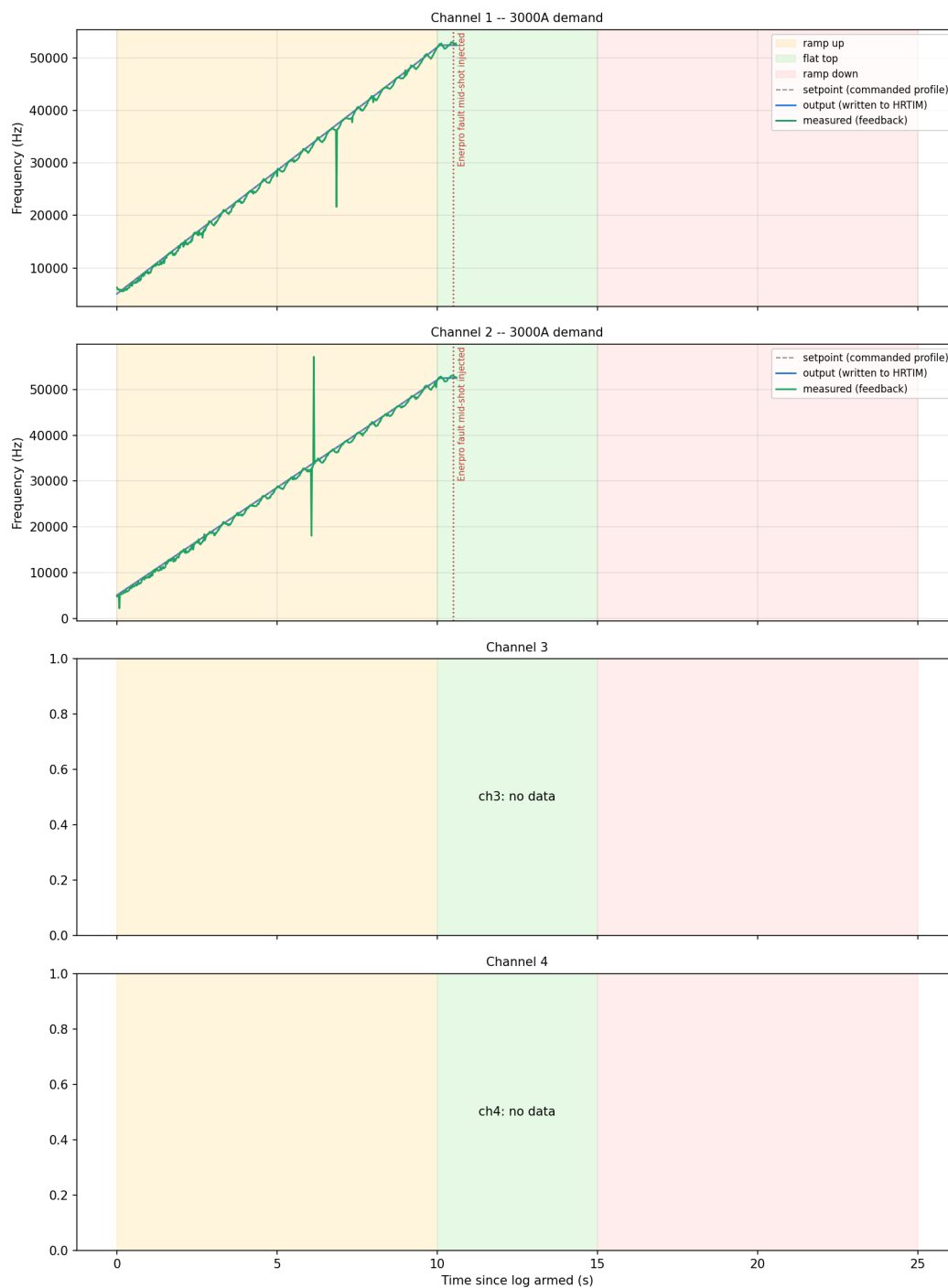
ocp -- OCP fault mid-shot
Verdict: PASS
STATE? -> 'OK FAULT OVERCURRENT 1'; survivor ch2 running=1 output=21660; faulted/survivor logged (584/584 samples).



3 Enerpro fault injection (enerpro) — PASS

Same 2-channel setup (10/5, 3000 A), with an Enerpro fault injected on channel 1 mid-shot. The test **passed** because STATE? reported ENERPRO 1 and the survivor derated and kept running (21 638 Hz), confirming Enerpro faults use the overcurrent-style *reduce-and-keep-running* response rather than a Water/Temp full stop — validating the 2026-09-18 fault reclassification.

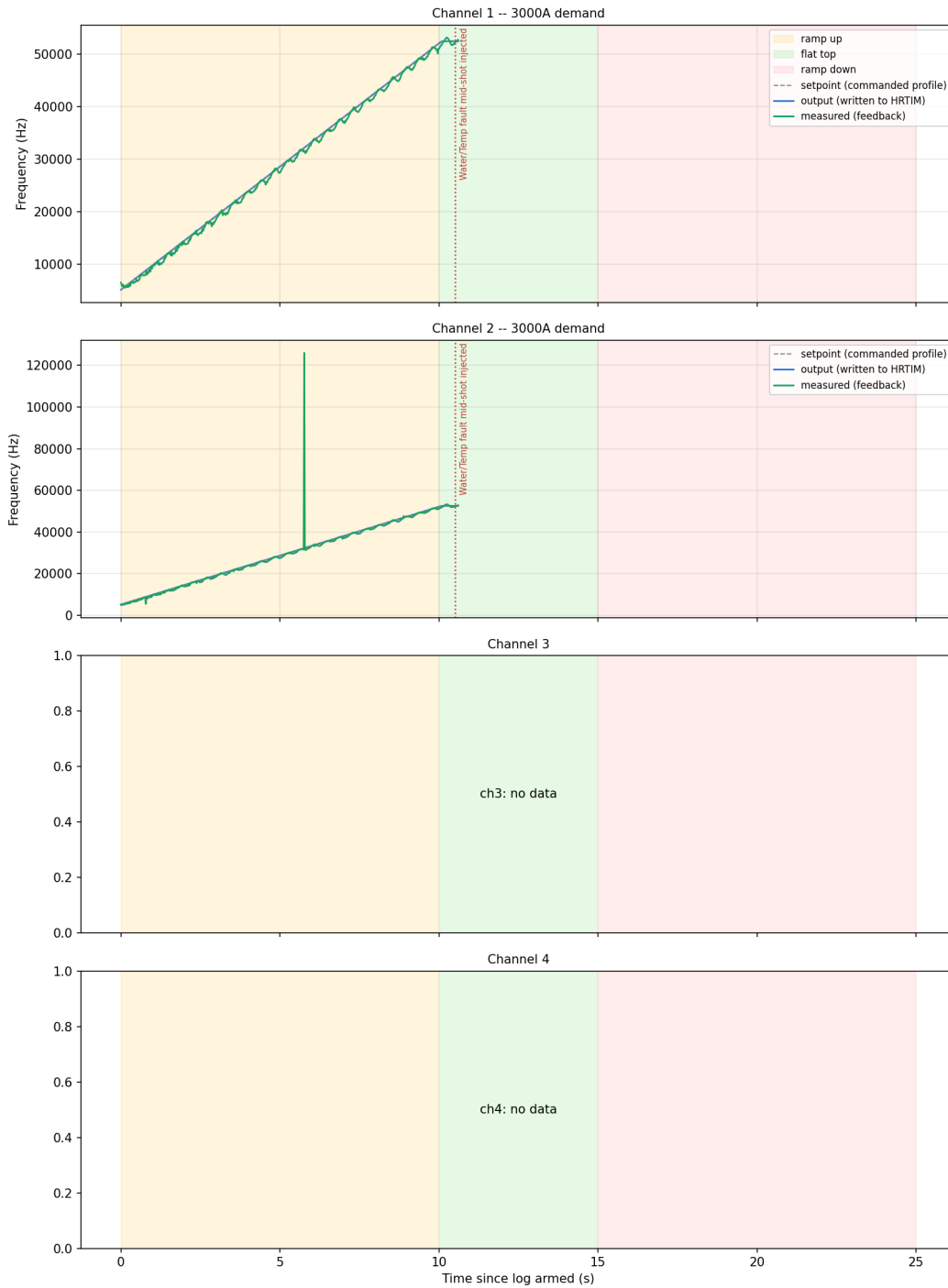
enerpro -- Enerpro fault mid-shot
Verdict: PASS
STATE? -> 'OK FAULT ENERPRO 1'; survivor ch2 running=1 output=21638; faulted/survivor logged (584/584 samples).



4 Water/Temp fault injection (watertemp) — PASS

Same 2-channel setup (10/5, 3000 A), with a combined Water+Temp fault injected on channel 1 mid-shot. The test **passed** because STATE? reported GENERAL — the distinct full-stop fault type — confirming that Water/Temp trips the system-wide open-loop ramp-down to 0 A on every active channel, rather than the derate-and-survive behavior of OCP/Enerpro.

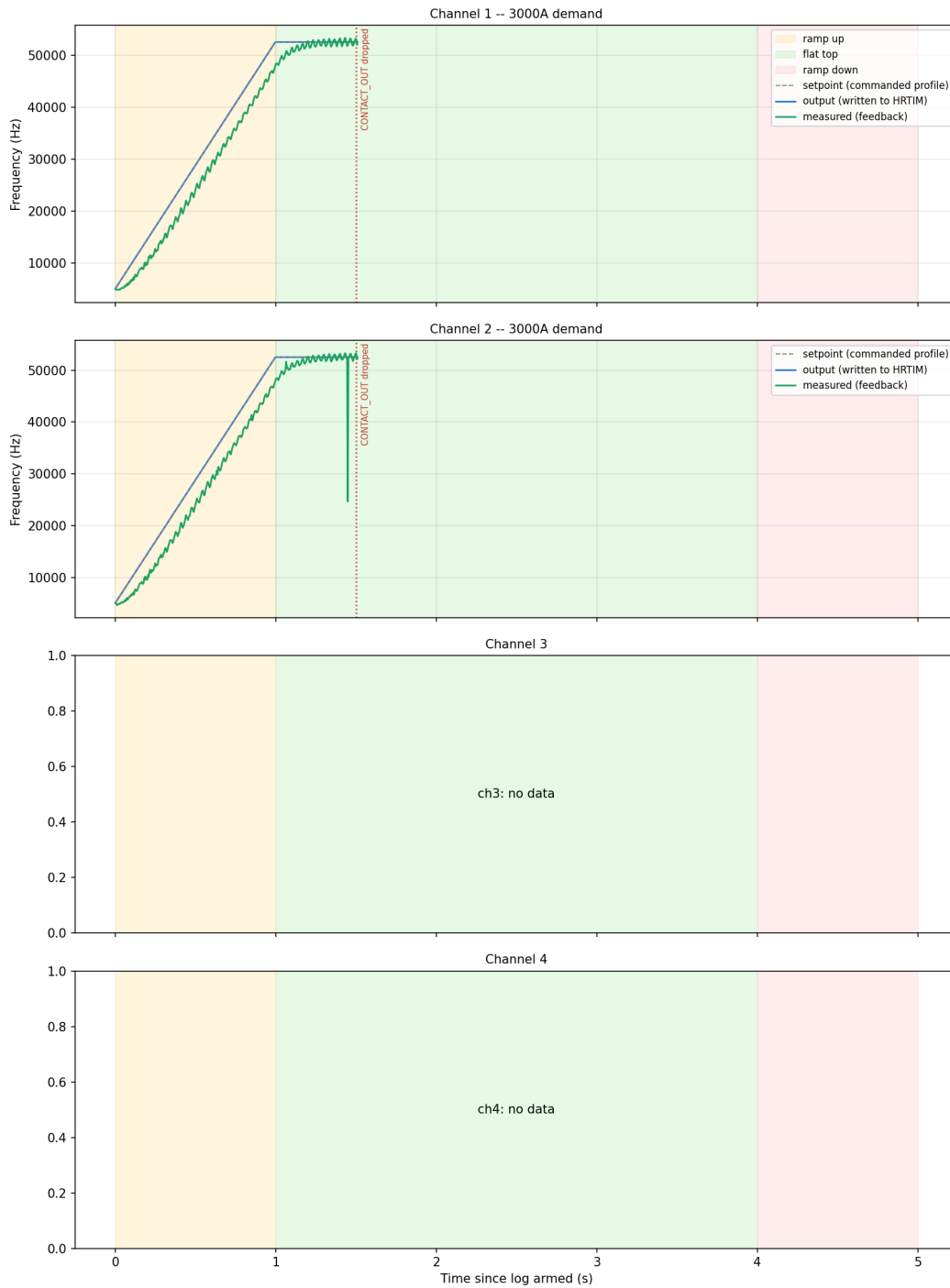
watertemp -- Water/Temp fault mid-shot
Verdict: PASS
STATE? -> 'OK FAULT GENERAL'; survivor ch2 running=1 output=42240; faulted/survivor logged (584/584 samples).



5 ENA_OUT/CONTACT_OUT drop (ena_contact) — PASS

Two channels were gated (10/5, 3000 A) and the controller's own CONTACT_OUT for channel 1 was dropped mid-shot. The test **passed** because STATE? reported ENABLE_OUTPUT 1 and the survivor derated to 21 660 Hz — validating the per-channel self-commanded-output fault (SM_FAULT_ENABLE_OUTPUT), which is distinct from the external-enable *input* interlock.

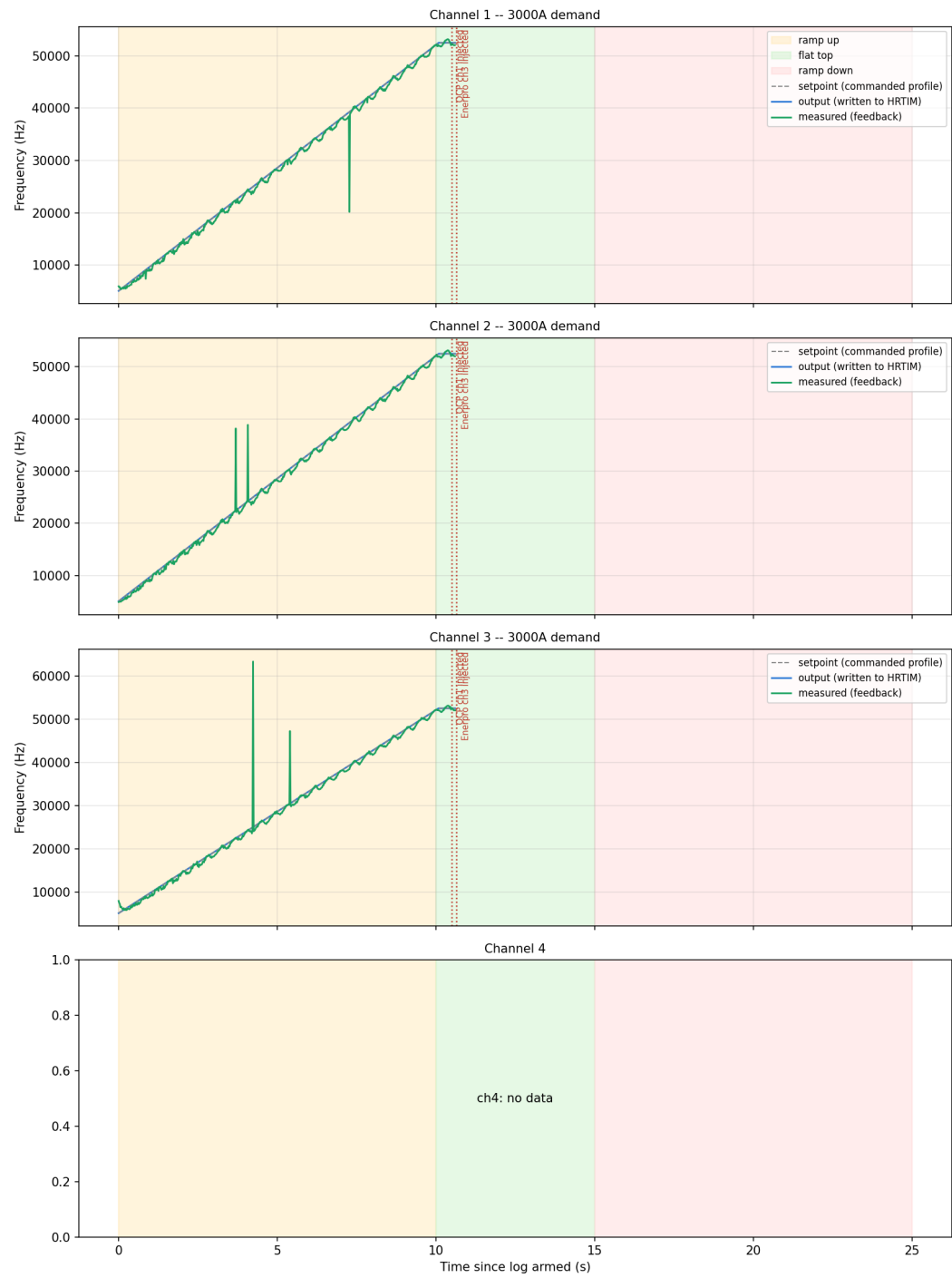
ena_contact -- ENA_OUT/CONTACT_OUT dropped mid-shot
Verdict: PASS
STATE? -> 'OK FAULT ENABLE_OUTPUT 1'; survivor running=1 output=21660



6 Fault sequencing (multi_fault) — PASS

Three channels were gated (10/5, 3000 A) and two different faults were injected in sequence — OCP on channel 1, then Enerpro on channel 3 — with channel 2 as the survivor. The test **passed** because both faulted channels ended disabled, STATE? preserved the *first* fault's identity (OVERCURRENT 1) even after the second fault, and the survivor visibly derated after each fault (29 989 Hz → 24 769 Hz) — confirming the “already-faulted” short-circuit still hard-disables the newly-faulted channel without clobbering the reported fault type.

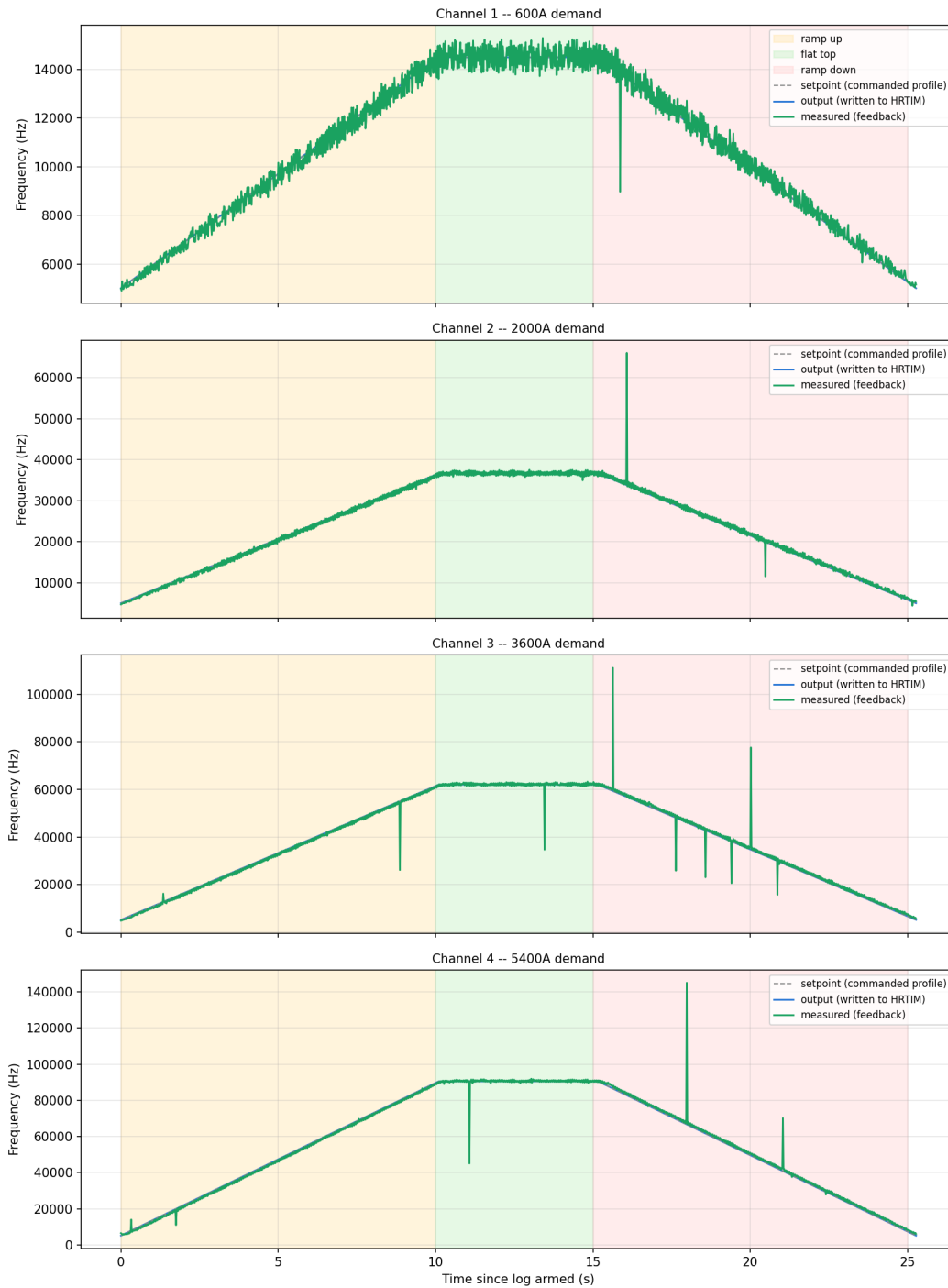
multi_fault -- Multi-fault sequencing (OCP then Enerpro, 2 channels)
Verdict: PASS
LT OVERCURRENT 1': survivor running=1 output=29989; after 2nd ('OK FAULT OVERCURRENT 1'): survivor running=1 output=24769; ch1 enabled='OK 0'; ch



7 Four channels simultaneously (all_channels) — PASS

All four channels were gated and fired at once (10/5) at per-channel demands of 600 / 2000 / 3600 / 5400 A. The test **passed** because every channel's flat-top median measured — output error stayed well under the 500 Hz threshold (377 / 387 / 362 / 363 Hz), confirming four independent Transrex channels run and measure simultaneously with no cross-channel interference.

all_channels -- All 4 channels gated and firing simultaneously (600/2000/3600/5400 A)
 Verdict: PASS
 output median error=377Hz; ch2 flat-top measured_vs_output median error=387Hz; ch3 flat-top measured_vs_output median error=362Hz; ch4 flat-top measured_vs_output median error=362Hz



8 Channel/demand matrix sweep — 45/45 PASS

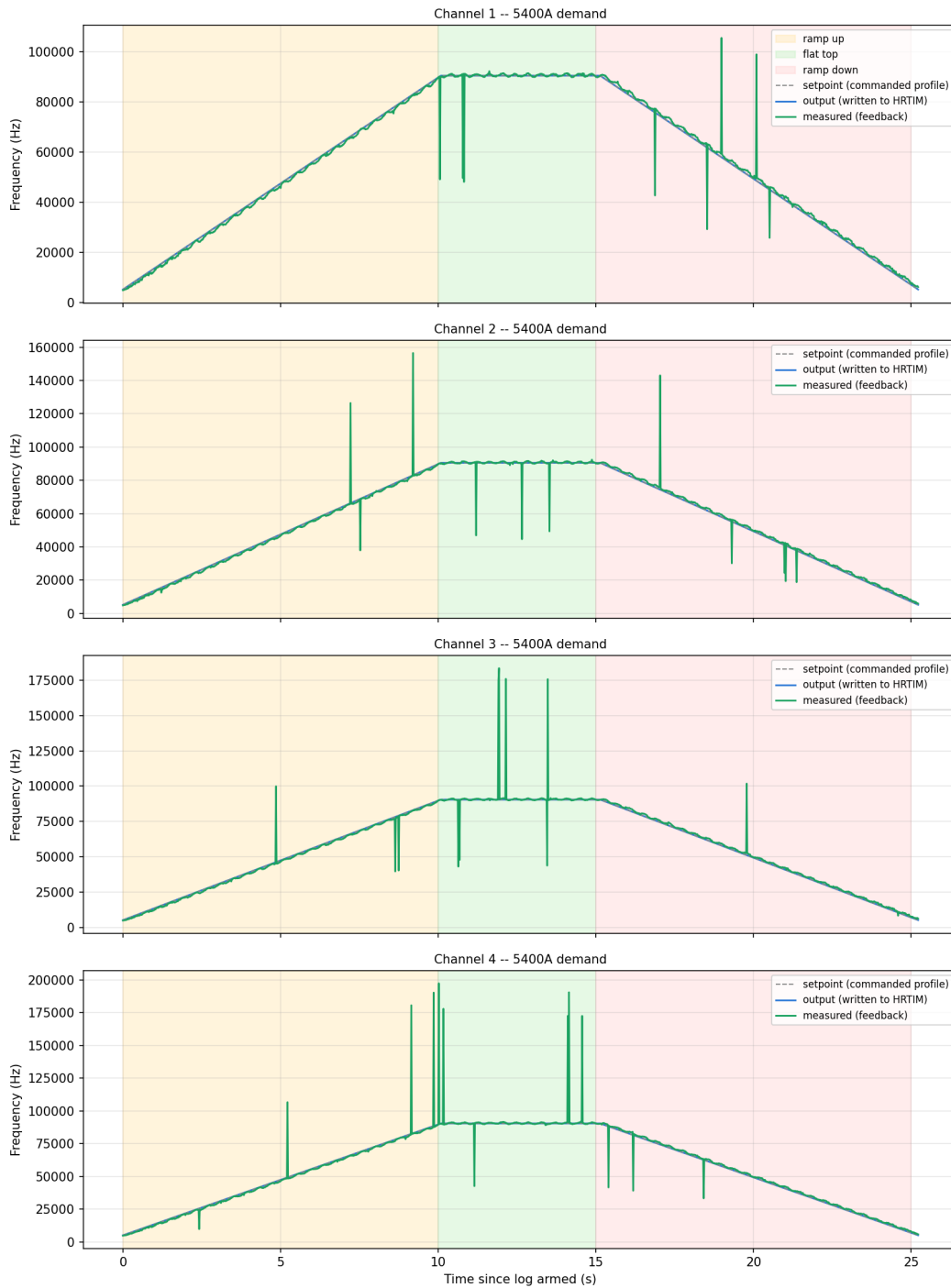
The stress sweep ran 45 open-loop shots (10/5) — one per non-empty subset of $\{1, 2, 3, 4\}$ at each of three demand levels (600 / 3000 / 5400 A = 10/50/90%). Every one of the 45 shots **passed** with a worst output — setpoint error of **0 Hz**, confirming the profile generator, per-channel HRTIM write, and the fiber feedback round-trip are clean across the full channel-combination $\times$ operating-point envelope with no cross-channel coupling.

Channels	Demand (A)	Worst error (Hz)	Verdict
1	600	0	PASS
1	3000	0	PASS
1	5400	0	PASS
2	600	0	PASS
2	3000	0	PASS
2	5400	0	PASS
3	600	0	PASS
3	3000	0	PASS
3	5400	0	PASS
4	600	0	PASS
4	3000	0	PASS
4	5400	0	PASS
1+2	600	0	PASS
1+2	3000	0	PASS
1+2	5400	0	PASS
1+3	600	0	PASS
1+3	3000	0	PASS
1+3	5400	0	PASS
1+4	600	0	PASS
1+4	3000	0	PASS
1+4	5400	0	PASS
2+3	600	0	PASS
2+3	3000	0	PASS
2+3	5400	0	PASS
2+4	600	0	PASS
2+4	3000	0	PASS
2+4	5400	0	PASS
3+4	600	0	PASS
3+4	3000	0	PASS
3+4	5400	0	PASS
1+2+3	600	0	PASS
1+2+3	3000	0	PASS
1+2+3	5400	0	PASS
1+2+4	600	0	PASS
1+2+4	3000	0	PASS
1+2+4	5400	0	PASS
1+3+4	600	0	PASS
1+3+4	3000	0	PASS

Channels	Demand (A)	Worst error (Hz)	Verdict
1+3+4	5400	0	PASS
2+3+4	600	0	PASS
2+3+4	3000	0	PASS
2+3+4	5400	0	PASS
1+2+3+4	600	0	PASS
1+2+3+4	3000	0	PASS
1+2+3+4	5400	0	PASS

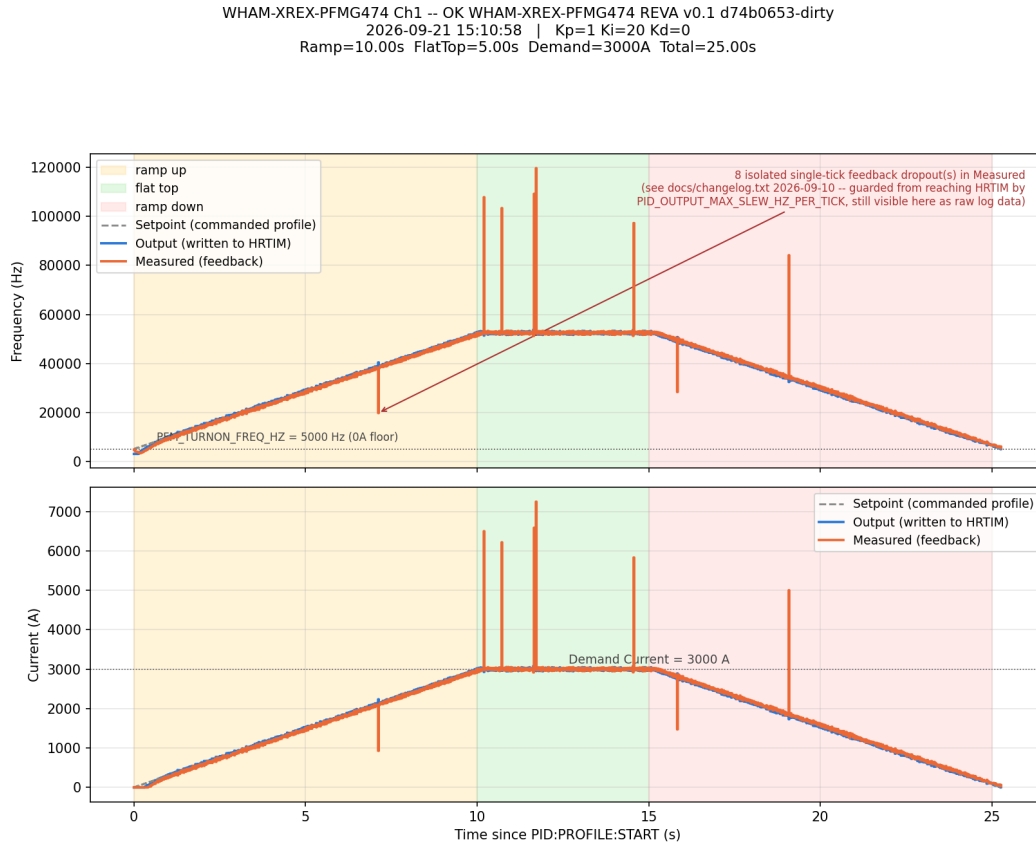
Representative plot (all four channels at the 90 % demand level):

matrix_1+2+3+4_5400A -- All 4 channels, 5400 A demand
Verdict: PASS
worst channel output_error max_abs=0 Hz



9 Closed-loop profiled shot — PASS (smoke test)

A one-shot closed-loop counterpart to the profiled shot: channel 1 gated through the simulator, the PID in **closed-loop** mode with PI gains ($K_p = 1.0$, $K_i = 20$, $K_d = 0$), and the same 10 s/5 s profile commanded (3000 A). Unlike the open-loop runs above, the PID must actually *drive* the output so the measured feedback tracks the profile. It **passed**: the flat-top median measured – setpoint error was **393 Hz** (1315 samples @ 52 Hz, full 25 s shot) — the loop holds the feedback to within the simulator’s own ripple noise of the commanded profile. (The 67 218 Hz maximum is the known single-sample feedback glitch, which the median is robust to.)



Notes

- The gain_sweep ki100 FAIL is the final-sample criterion, not a control defect (§1); a median-of-last-0.5 s verdict would pass it.
- Fault tests plot the injection instant as a dashed red line + label on every populated panel.
- Raw per-channel CSVs and all plots are under `shots/20260921_14*_*/`.